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8.2 Studi Kasus

1. Buatlah desain database untuk suatu perusahaan:
 - a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.
 - b. Table employee dengan constraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_name NOT NULL, email, salary, commission_pct, hire_date NOT NULL>
 - c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).
2. Analisis table-table dengan compute statistics
3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).
4. Analisis table-table dengan compute statistics
5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.
6. Kemudian tampilkan struktur dari view empvu80.
7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

-- Tabel Department

```
CREATE TABLE departments (  
    department_id INT PRIMARY KEY,  
    department_name VARCHAR(50) NOT NULL,  
    manager_id INT,  
    location_id INT  
);
```

-- Tabel Employee

```
CREATE TABLE employees (  
    employee_id INT PRIMARY KEY,  
    last_name VARCHAR(50) NOT NULL,  
    email VARCHAR(100),  
    salary DECIMAL(10,2),  
    commission_pct DECIMAL(5,2),
```

```
hire_date DATE NOT NULL,  
department_id INT,  
CONSTRAINT fk_dept  
    FOREIGN KEY (department_id)  
    REFERENCES departments(department_id)  
);
```

-- Tabel Tambahan: Suppliers

```
CREATE TABLE suppliers (  
    supplier_id INT PRIMARY KEY,  
    supplier_name VARCHAR(100) NOT NULL,  
    contact_name VARCHAR(50)  
);
```

-- Tabel Tambahan: Inventory

```
CREATE TABLE inventory (  
    item_id INT PRIMARY KEY,  
    item_name VARCHAR(100),  
    stock_quantity INT,  
    supplier_id INT,  
CONSTRAINT fk_supplier  
    FOREIGN KEY (supplier_id)  
    REFERENCES suppliers(supplier_id)  
);
```

-- Compute Statistics

```
ANALYZE TABLE departments COMPUTE STATISTICS;
```

ANALYZE TABLE employees COMPUTE STATISTICS;

ANALYZE TABLE suppliers COMPUTE STATISTICS;

ANALYZE TABLE inventory COMPUTE STATISTICS;

-- Insert Data Department

INSERT INTO departments VALUES (10, 'Administration', 200, 1700);

INSERT INTO departments VALUES (20, 'Marketing', 201, 1800);

INSERT INTO departments VALUES (30, 'Purchasing', 202, 1700);

INSERT INTO departments VALUES (40, 'Human Resources', 203, 2400);

INSERT INTO departments VALUES (50, 'Shipping', 204, 1500);

INSERT INTO departments VALUES (60, 'IT', 205, 1400);

INSERT INTO departments VALUES (70, 'Public Relations', 206, 2700);

INSERT INTO departments VALUES (80, 'Sales', 207, 2500);

INSERT INTO departments VALUES (90, 'Executive', 208, 1700);

INSERT INTO departments VALUES (100, 'Finance', 209, 1700);

-- Insert Data Employee

INSERT INTO employees

(employee_id, last_name, email, salary, commission_pct, hire_date,
department_id)

VALUES

(1, 'King', 'king@mail.com', 5000, 0.10, '2023-01-01', 90),

(2, 'Kochhar', 'koch@mail.com', 4500, NULL, '2023-02-15', 90),

(3, 'Abel', 'abel@mail.com', 3000, 0.20, '2023-03-10', 80),

(4, 'Taylor', 'taylor@mail.com', 3200, 0.20, '2023-04-05', 80),

(5, 'Grant', 'grant@mail.com', 2800, 0.15, '2023-05-12', 80),

(6, 'Whalen', 'whalen@mail.com', 4000, NULL, '2023-06-01', 10),

(7, 'Zlotkey', 'zlot@mail.com', 3100, 0.20, '2023-07-20', 80),

```
(8, 'Hunold', 'hunold@mail.com', 4200, NULL, '2023-08-14', 60),  
(9, 'Ernst', 'ernst@mail.com', 3900, NULL, '2023-09-05', 60),  
(10, 'Lorentz', 'lor@mail.com', 2500, 0.10, '2023-10-10', 80);
```

```
-- Membuat View empvu80
```

```
CREATE VIEW empvu80 AS
```

```
SELECT
```

```
    employee_id AS id_number,
```

```
    last_name AS name,
```

```
    salary,
```

```
    department_id
```

```
FROM employees
```

```
WHERE department_id = 80;
```

```
-- Menampilkan Struktur View
```

```
DESCRIBE empvu80;
```

```
-- Hapus Index Lama (jika ada)
```

```
DROP INDEX emp_dept_id_idx;
```

```
-- Membuat Non-Unique Index pada Foreign Key
```

```
CREATE INDEX emp_dept_id_idx
```

```
ON employees(department_id);
```

Script Output x



| Task completed in 0.038 seconds

Name	Null?	Type
ID_NUMBER	NOT NULL	NUMBER(38)
NAME	NOT NULL	VARCHAR2(50)
SALARY		NUMBER(10,2)
DEPARTMENT_ID		NUMBER(38)